FIITJEE

JEE(Main)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
CONCEPT RECAPITULATION TEST – III
(Main)

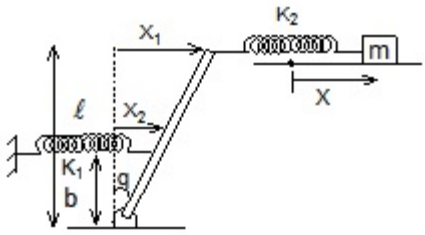
Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**B**	31.	**B**	61.	**D**
2.	**D**	32.	**D**	62.	**B**
3.	**C**	33.	**B**	63.	**B**
4.	**C**	34.	**C**	64.	**C**
5.	**A**	35.	**B**	65.	**B**
6.	**C**	36.	**A**	66.	**B**
7.	**C**	37.	**C**	67.	**B**
8.	**C**	38.	**A**	68.	**D**
9.	**A**	39.	**C**	69.	**A**
10.	**D**	40.	**B**	70.	**A**
11.	**B**	41.	**A**	71.	**A**
12.	**D**	42.	**B**	72.	**A**
13.	**A**	43.	**A**	73.	**A**
14.	**C**	44.	**A**	74.	**A**
15.	**B**	45.	**A**	75.	**B**
16.	**B**	46.	**C**	76.	**B**
17.	**D**	47.	**D**	77.	**D**
18.	**C**	48.	**A**	78.	**C**
19.	**C**	49.	**A**	79.	**A**
20.	**D**	50.	**B**	80.	**A**
21.	**C**	51.	**B**	81.	**B**
22.	**C**	52.	**A**	82.	**C**
23.	**D**	53.	**B**	83.	**A**
24.	**A**	54.	**B**	84.	**A**
25.	**D**	55.	**A**	85.	**D**
26.	**B**	56.	**C**	86.	**B**
27.	**B**	57.	**C**	87.	**C**
28.	**B**	58.	**C**	88.	**A**
29.	**B**	59.	**A**	89.	**A**
30.	**B**	60.	**A**	90.	**C**

Physics

PART – I

SECTION – A

1. $\frac{X_2}{b} = \frac{X_1}{\ell} \longrightarrow (1)$



Torque equation about Hinge

$K_2(x - x_1).\ell = K_1 x_2 . b \longrightarrow (2)$

Substituting X_2 from equation (1) & on Solving we get

$X_1 = \frac{K_2 L^2 x}{\left[K_2 L^2 + K_1 b^2\right]}$

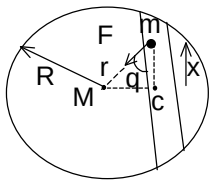
Equation of mass $F = ma = -K_2(x - x_1)$

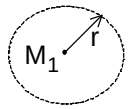
On solving = $a = \frac{-K_1 K_2 b^2 x^2}{m\left(K_2 L^2 + K_1 b^2\right)}$

2. $T = \frac{T_{Left}}{2} + \frac{T_{Right}}{2} = \pi\sqrt{\frac{m}{K_1 + K_2}} + \pi\sqrt{\frac{m}{K_1}}$

Note string becomes slack when mass pushes the string

3. $F_{net/c} = \frac{-GmM^1}{r^2}\cos\theta$





$ma = -\frac{GmM}{\frac{4\pi R^3}{3}}\frac{4\pi}{3}.\frac{r^3 x}{r^2 r}$

$a = -\frac{GmM}{mR^3}x \quad or\ \omega = \sqrt{\frac{GM}{R^3}} = \sqrt{\frac{g}{R}}$

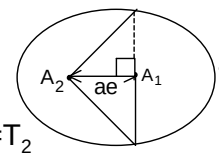
$\therefore$ T.P. in any earth tunnel is $T = 2\pi\sqrt{R/g}$

4. $E = \frac{1}{2}KA^2.-(bt/m)$

$\therefore \frac{1}{4} = \frac{E(t=t)}{E(t=0)} = e^{-bt/m}$

Taking log and solving we get $t = \frac{2 \times 0.693}{b/m} = 6.93\,\text{sec}$

5. $\because \frac{dA}{dt} = c\left[kepler's\ 2nd\ law\right]$



$T_1 = T_{ACB}$

$T_{BDA} = T_2$

$\therefore \frac{T_1}{T_2} = \frac{A_1}{A_2} = \frac{\frac{\pi ab}{2} + \frac{1}{2} \times 2b \times ae}{\frac{\pi ab}{2} - \frac{1}{2} \times 2b \times ae}$

$or\ \frac{T_1}{T_2} = \frac{(\pi + 2e)}{(\pi - 2e)}$

6. $\frac{mv_E^2}{2}\frac{-3}{2}\frac{GmM}{R}=0\left[for\ escape\ velocity\ V_E\right]$

$V_E=\sqrt{3gR}$

For S.H.M in tunnel $\left[\omega=\sqrt{g/R}\right]$

$V_{Max}=\omega A=\left(\sqrt{\frac{g}{R}}\right)\frac{R}{2}=\frac{\sqrt{gR}}{2}$

7. $N_A=N_B=\frac{GmM}{R^2}-m\omega^2R-\frac{mv^2}{R}$

$\left(\frac{GmM}{R^2}-m\omega^2R-N_A=\frac{mv^2}{R}\right)$

8. $F_B=\sqrt{F_{B/x}{}^2+F_{B/y}{}^2}$

$F_B=\sqrt{\left[V\rho_\ell\left(g+a_y\right)\right]^2+\left[V\rho_\ell a_x\right]^2}$

$=10\times10^{-4}\times10^3\sqrt{(12)^2+(9)^2}=15N$

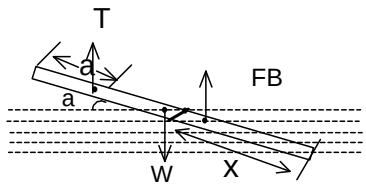
9. $\tau_{net}=0\left[about\ point\ of\ suspension\right]$

$F_B\left(\ell-a-\frac{x}{2}\right)\cos\alpha=w\left(\frac{\ell}{2}-a\right)\cos\alpha$

Putting $F_B=Ax\rho_\ell$ & $W=A\ell\rho_s$

$x=(\ell-a)\pm\sqrt{(\ell-a)^2-\frac{\rho_\ell}{\rho_0}\ell(\ell-2a)}=0.1m$

Since x < (ℓ - a), discard '+' sign in the solution



10. Momentum of water following out of the tube in time $\Delta t = \Delta p = \rho Av^2\Delta t$ where $v=\sqrt{2gH}$

$\therefore F\Delta t=\Delta p=\rho Av^2\Delta t$

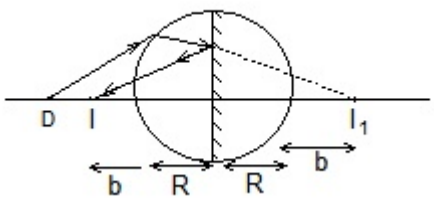
$or\ F=\rho Av^2=2\rho gHA$

$\therefore$ Reading of spring balance = $W_0-2\rho g\,HA$

11. $P=P_{liquid\ surface}-\rho gh$

= $0-\rho gh$ = -2000 N / m^2

12. From symmetry $I_1I_2=2(b+R)=2(6+5)=22cm$



13. $\sin\alpha=\frac{1}{\mu g}$

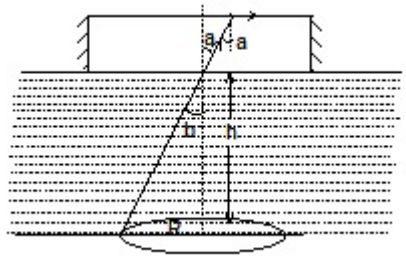
$\mu_w\sin\beta=\mu_g\sin\alpha$

$or\ \mu_w \sin\beta = \mu_g \times \frac{1}{\mu_g} = 1$

$\beta = \sin^{-1}\left(\frac{1}{\mu_w}\right)$

$\therefore A_{base} = \pi R^2 = \pi h^2 \tan^2\beta = \frac{\pi h^2}{\left(\mu_w{}^2 - 1\right)}\left[\because \tan\beta = \frac{1}{\sqrt{\mu_w{}^2 - 1}}\right]$

$= 9\pi.m^2$



14. $I_{3R} = \left(-\frac{\sqrt{3}R}{2}, \frac{R}{2}\right) \& \left(\frac{+R}{2}, +\frac{\sqrt{3}R}{2}\right)$

15. $\sin(90-\theta) = \frac{1}{\mu_1} \Rightarrow \cos\theta = \frac{1}{\mu_1} \longrightarrow (1)$

$\sin(90-2\theta) = \frac{1}{(\mu_1/\mu_2)}$

$\cos 2\theta = \frac{\mu_2}{\mu_1} = \frac{\mu_2}{(1/\cos\theta)} \therefore \mu_2 = \frac{\cos 2\theta}{\cos\theta}$

16. Dimensional analysis

17. $I = \left(\frac{I_0}{2}\right)\cos^2\theta.\sin^2\theta = \frac{I_0}{2}\left[\frac{\sin 2\theta}{2}\right]^2 = \frac{I_0}{8}\sin^2 2\theta$

18. F.O.V. = 1 mm

$F.W. = \frac{D\lambda}{2d} = 0.25mm$

19. $I = 2\pi^2\rho v A^2 \upsilon^2 = \frac{W}{4\pi r^2}$

$\therefore A = \left[\frac{W}{8\pi^3\rho v \upsilon^2}\right]^{1/2} . \frac{1}{r} \cong 10^{-6} m$

20. $2T\sin\theta = dm\omega^2 R$

$\approx 2T\theta = \mu d\ell\omega^2 R$

$\approx 2T\theta = \mu R^2\theta\omega^2 R$

$\Rightarrow \frac{T}{\mu} = \omega^2 R^2$

$or\ v_{wave} = \omega R$

Note v_{wave} obtained is always with respect to medium

$\therefore v_{wave/ground} = \omega R - \omega R$

$= 0$

21. Number of waves received by observation = f

Time taken by observer to receive all wave

$$\frac{Tfo}{\left(\frac{\Delta x}{v_{observer}}\right)} = f \longrightarrow (1)$$

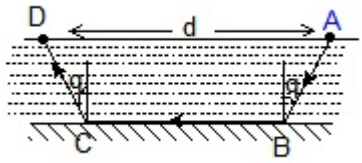
$[T = duration\ of\ beep]$

Also $f = \left(\frac{c + v_{obs}}{c}\right) fo \longrightarrow$(ii)

On solving (i) & (ii) we get

$$T = \frac{f\Delta x}{c(f - fo)} = 5\,\text{sec}\,onds$$

22. Method 1



$$T = t_{AB} + t_{BC} + t_{CD}$$

$$T = \frac{2D}{\cos\theta V_w} + \frac{(d - 2D\tan\theta)}{v_R}$$

$$\frac{dT}{d\theta} = 0 \Rightarrow \frac{2D\sec\theta\tan\theta}{v_w} - \frac{2D\sec^2\theta}{v_R}$$

$$\Rightarrow \sin\theta = \frac{v_w}{v_R}$$

$$\therefore T = \frac{2D}{\sqrt{1 - \frac{v_w^2}{v_R^2}}.v_w} + \frac{\left(d - 2D\frac{v_w}{\sqrt{v_R^2 - v_w^2}}\right)}{V_R}$$

$$T = \frac{dv_w + 2D\sqrt{v_r^2 - v_w^2}}{v_r v_w}$$

Method 2

Assume critical angle deviation like in light $\therefore \sin\theta\ \frac{1}{\mu R} = \frac{v_w}{v_R}$

Then solve $T = t_{AB} + t_{BC} + t_{CD}$

23. $$\frac{hc}{\lambda} - \frac{hc}{\lambda_0} = eEd$$

On solving we get $\frac{1}{\lambda_0} = \frac{1}{\lambda} - \frac{eEd}{hc}$

24. $$\left|\vec{F}\right| = \frac{2P\cos\theta}{c} [\perp to\ mirror\ \&\ directed\ toward\ it]$$

25. $\overset{v}{\bullet\!\!\rightarrow}$
2m

Before collision

$$pi = 2mv = \sqrt{4mk.E} = \sqrt{4m}\ units$$

$$p_1y = p_2y = \sqrt{2 \times m \times 1} = \sqrt{2m}\ units$$

$$= \left[\frac{p_2^2}{2m} + \frac{p_{1y^2}}{2m} - 1\right] ev$$

Energy of dissociation = $= \left[\frac{\left(\sqrt{4m}\right)^2 + \left(\sqrt{2m}\right)^2}{2m} + \frac{\left(\sqrt{2m}\right)^2}{2m} - 1\right] ev$

$$= [3 + 1 - 1] ev$$

$$= 3ev$$

26. $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} \quad \therefore R = 2\Omega$

$\frac{dR}{R^2} = \frac{dR_1}{R_1^2} + \frac{dR_2}{R_2^2} \quad \Rightarrow dR = 0.05\Omega$

$\therefore R = (2 \pm 0.05)\Omega$

27. $d = 20.1\, mm\,[3\, significant\ figure]$

$v = \frac{4\pi}{3}\left(\frac{d}{2}\right)^3 = \frac{\pi d^3}{6} = 4.253648 cm^3$

Taking 3 significant figures only, we get
V = 4.25 cm^3

28. $i_{zener} = 5 i_{load} = 20mA \ \& \ i_{RS} = i_{zener} + i_{Load} = 24mA$

$V_{Rs} + V_{zener} = 11$

$V_{RS} + 7 = 11$

$V_{RS} = 4V$

$\therefore R_s = \frac{V_{Rs}}{i_{Rs}} = \frac{4}{24 \times 10^{-3}}\Omega = 167\Omega$

29. Consider a case of an n type semi-conductor. The majority carries density (n) is considerably larger than the minority hole density $p\,(i.e.\ \mathrm{n} >> \mathrm{p})$.On illumination, $n_f = n + \Delta n \ \& \ p_f = p + \Delta p$

Since $\Delta n = \Delta p$ {equal hole electron pair generation in bond breaking}

$\frac{\Delta n}{n} << \frac{\Delta p}{p}$

$\therefore$ It is difficult to measure small $\frac{\Delta n}{n}$

Compared to $\Delta p / p$

$\therefore$ Photodiode is used in Reverse bios mode since it is easier to measure the higher fractional change of minority carriers

30. $d = \sqrt{2Rh_T} + \sqrt{2Rh_R}$

$= 288 \times 10^2 \times \sqrt{10} = 91km$

Chemistry

PART – II

SECTION – A

31. Photochemical smog has high concentration of oxidizing agents.

32. Bithionol / Bithional is added to the toilet soap to remove the bad odour produced by bacterial decomposition of organic matter on the skin.

33.
$$\underset{a\ mole}{2Fe} + \underset{\frac{a}{2}mole}{O_2} \longrightarrow \underset{a}{2FeO}$$

$$\underset{b}{4Fe} + \underset{\frac{3b}{4}}{3O_2} \longrightarrow \underset{\frac{b}{2}}{2Fe_2O_3}$$

$\therefore a + b = 1$

$\frac{a}{2} + \frac{3b}{4} = 0.65$

$\therefore a = 0.4\ mole$

$b = 0.6\ mole$

$\therefore moles\ of\ FeO = 0.4$

$moles\ of\ Fe_2O_3 = 0.3$

$\therefore ratio = 4:3$

34. gold number = 0.03
Gold sol volume = 100 ml
Let the amount of hemoglobin required is x mg
Since the gold number is the amount in miligrams required to prevent the coagulation of 10 ml of gold sol by 1 ml of 10% NaCl

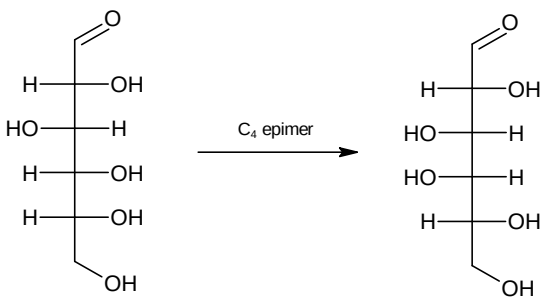
$\therefore \frac{x}{100} \times 10 = gold\ number$

$or\ \frac{x}{10} = 0.03$

$\therefore x = 0.30\ mg$

35. Wood-pulp, on dissolving in CS_2-- - alkali solution forms sodium salt of cellulose xanthate not the rayon.

36.



D-Glucose D-Galactose

37. $IO_3^- + 5I^- + 6H^+ \longrightarrow 3I_2(s) + 3H_2O$

38. $PM = \rho RT$

$or\ \rho T = \dfrac{PM}{R}$

$or\ \rho T = constant$

39. $QV = \dfrac{1}{2}mv^2$

$\therefore v = \sqrt{\dfrac{2QV}{m}}$

$\therefore v_P = \sqrt{\dfrac{2QV}{m_P}}\ or\ v_\alpha = \sqrt{\dfrac{2\times 2QV}{m_\alpha}}$

$m_P.v_P = \sqrt{2QV.m_P}\ \&\ m_\alpha.v_\alpha = \sqrt{4QV.m_\alpha}$

$\therefore ratio = \sqrt{\dfrac{m_P}{2m_\alpha}}$

40. Let x and y are the concentration of Br^- and Cl^- at equilibrium when $E_{cell} = 0$

$\therefore \left[Ag^+\right]_{LHS} = \dfrac{K_{sp}(ArBr)}{\left[Br^-\right]} = \dfrac{5\times 10^{-13}}{x}$

$\left[Ag^+\right]_{RHS} = \dfrac{K_{sp}(AgCl)}{\left[Cl^-\right]} = \dfrac{1\times 10^{-10}}{y}$

Also $E_{cell} = E^o_{Ag/Ag^+}(RHS) + E^o_{Ag/Ag^+}(LHS) + \dfrac{0.0591}{1}\log\dfrac{\left[Ag^+\right]_{RHS}}{\left[Ag^+\right]_{LHS}}$

$0 = 0 + \dfrac{0.0591}{1}\log\dfrac{1\times 10^{-10}\times x}{5\times 10^{-13}\times y}$

$\therefore \dfrac{x}{y} = \dfrac{1}{200}$

41. $\Delta S_V = \dfrac{RT_bM}{1000K_b}$

42. Decomposition of HI on Au surface is zero order reaction

$\therefore P_{HI} = -Kt + P^o_{HI}$

P_{H_2} increases continuously $2HI \xrightarrow{Au\ surface} H_2 + I_2$

43. For cyclic process, du = 0

as, q = -ve

$\therefore$ work is done by the system

$\therefore$ q = - w = area occupied by circle

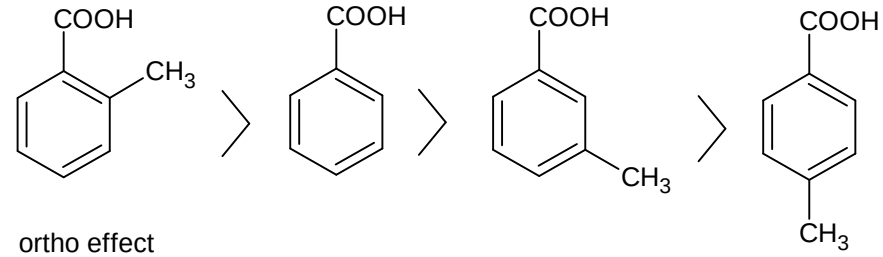
$= \pi r^2 = \dfrac{22}{7}\times 10^2$

44. A

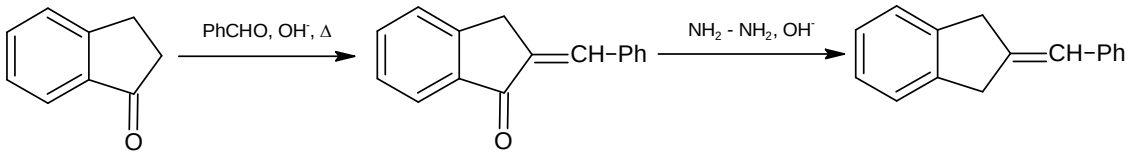
45. $CaSO_4.2H_2O$, $2CaO.3Ba_2O_3.5H_2O$, CaH_2 are ores of Ca.

46. $D_2 > H_2 > F_2$ is the order of BDE.

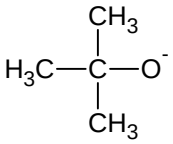
47. D

48. K_a order = 

ortho effect

49. 

50. Hofmann rearrangement occurs without altering the stereochemistry.

51.  is bulkier base

52. Nitrous acid reaction with 2° amine.

53. MnO_2 oxidizes allylic | benzylic 1° | 2° alcohols while
Jones reagent oxidizes 1° | 2° alcohols with out affecting C = C bond | C ≡ C bond.

54. reactivity of – Br is higher than – Cl for RMgX formation.

55. for the indicator $HIn \rightleftharpoons H^+ + In^-$
According to the question, 20% of the final form of the indicator is blue (acidic) and 80% red (basic)

$$K_{In} = \frac{\left[H^+\right]\left[In^-\right]}{\left[HIn\right]}$$

$$\left[H^+\right] = K_{In}\frac{\left[In^-\right]}{\left[HIn\right]} = 1\times10^{-5}\left(\frac{80}{20}\right) = 4\times10^{-5}$$

$pH = 4.4$

For 80% blue $\left[H^+\right] = 1\times10^{-5}\left(\frac{20}{80}\right)$

pH = 6 – 0.4 = 5.6

$\therefore$ change in pH = 5.6 – 4.4 = 1.2

56. C

57. SO_2 turns acidified $K_2Cr_2O_7$ solution green whereas CO_2 will have no effect on such a solution.

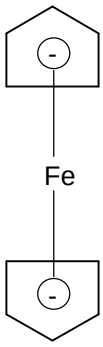
58. C

59. Nessler's reagent is a mixture of $K_2[HgI_4]$ and KOH.

$$2KI + HgCl_2 \longrightarrow 2KCl + HgI_2$$

$$HgI_2 + 2KI \longrightarrow K_2\left[HgI_4\right] \xrightarrow{+KOH} Nessler's\ reagent$$

60.



Mathematics

PART – III

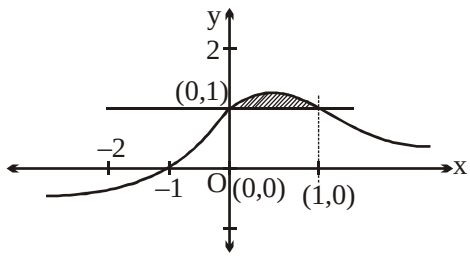
SECTION – A

61. Let $x^2 = 4\cos^2\theta + \sin^2\theta$
then $(4 - x^2) = 3\sin^2\theta$ and $(x^2 - 1) = 3\cos^2\theta$
$\therefore$ $f(x) = \sqrt{3}\,|\sin\theta| + \sqrt{3}\,|\cos\theta|$
$\Rightarrow$ $y_{min.} = \sqrt{3}$ and $y_{max.} = \sqrt{3}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right) = \sqrt{6}$.
Hence range of f(x) is $\left[\sqrt{3}, \sqrt{6}\right]$

62. By using condition of tangency, we get $4h^2 = 3k^2 + 2$
$\therefore$ Locus of P(h, k) is $4x^2 - 3y^2 = 2$ (which is hyperbola.)
Hence $e^2 = 1 + \frac{4}{3} \Rightarrow e = \sqrt{\frac{7}{3}}$

63. We have $|z|^2 + \frac{16}{|z|^3} = z^2 - 4z = \bar{z}^2 - 4\bar{z} \Rightarrow (z - \bar{z})(z + \bar{z} - 4) = 0$
$\Rightarrow z = \bar{z} = x\ (x \neq 2)$
So, $x^2 = 4x + x^2 + \frac{16}{|x|^3} \Rightarrow x = \frac{-4}{|x|^3} \Rightarrow x = -\sqrt{2}$
$\therefore z = -\sqrt{2}$
Hence only one z will satisfy above equation.]

64. On solving C : $y = \frac{x+1}{x^2+1}$ and the line L : y = 1,
we have
$\frac{x+1}{x^2+1} = 1 \Rightarrow x + 1 = x^2 + 1 \Rightarrow x(x-1) = 0 \Rightarrow x = 0, 1$
Thus, the points of intersection are at x = 0 and x = 1.



Required area = $\int_0^1 \left(\frac{x+1}{x^2+1} - 1\right) dx$ =
$\frac{1}{2}\int_0^1 \frac{2x}{x^2+1}dx + \int_0^1 \frac{1}{x^2+1}dx - \int_0^1 dx$
$= \left[\frac{1}{2}\ln(x^2+1) + (\tan^{-1}x) - x\right]_0^1 = \frac{1}{2}\ln 2 + \frac{\pi}{4} - 1$

65. Given $(\cot^{-1}x)(\tan^{-1}x) + \left(2 - \frac{\pi}{2}\right)\cot^{-1}x - 3\tan^{-1}x - 3\left(2 - \frac{\pi}{2}\right) > 0$
$\Rightarrow \cot^{-1}x\left(\tan^{-1}x + 2 - \frac{\pi}{2}\right) - 3\left(\tan^{-1}x + 2 - \frac{\pi}{2}\right) > 0$
$\left(As\ \tan^{-1}x - \frac{\pi}{2} = -\cot^{-1}x\right)$

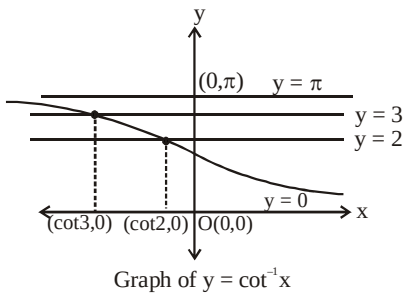
$\Rightarrow$ $(\cot^{-1}x - 3)(2 - \cot^{-1}x) > 0$

$\Rightarrow$ $(\cot^{-1}x - 3)(\cot^{-1}x - 2) < 0$

$\Rightarrow$ $2 < \cot^{-1}x < 3 \Rightarrow \cot 3 < x < \cot 2$

(As $\cot^{-1}x$ is a decreasing function.)

Hence $x \in (\cot 3, \cot 2)$]



Graph of $y = \cot^{-1}x$

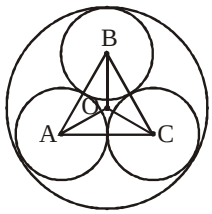
66. We have $(\tan x + \sec x) = 2\cos x \Rightarrow 1 + \sin x = 2\cos^2 x \Rightarrow 2\sin^2 x + \sin x - 1 = 0$

$\Rightarrow (2\sin x - 1)(\sin x + 1) = 0$

$\therefore \sin x = \frac{1}{2}$ or $\sin x = -1$ (Reject as $\sec x$ will not be defined.)

Hence $x = \frac{\pi}{6}, \frac{5\pi}{6}$.]

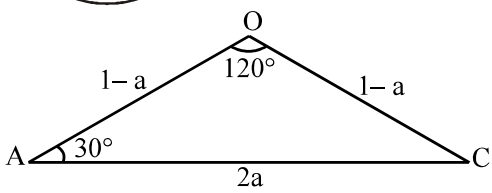
67. We have $\cos 30° = \frac{(2a)^2 + (1-a)^2 - (1-a)^2}{2(2a)(1-a)} = \frac{a}{1-a}$



$\Rightarrow \frac{\sqrt{3}}{2} = \frac{a}{1-a}$

$\Rightarrow$ $2a = \sqrt{3} - \sqrt{3}a \Rightarrow \left(2 + \sqrt{3}\right)a = \sqrt{3}$

$a = \frac{\sqrt{3}}{2+\sqrt{3}} = \sqrt{3}\left(2 - \sqrt{3}\right)$



68. For $x \geq 2$, L.H.S. is always non negative and R.H.S. is always – ve.
Hence for $x \geq 2$ no solution.
If $1 \leq x < 2$ then $(x - 2) = (x - 1) - 1 = x - 2$, which is an identity $\Rightarrow$ **(D)**
For $0 \leq x < 1$, LHS is '0' and RHS is (–) ve $\Rightarrow$ No solution.
For $x < 0$, LHS is (+) ve, RHS is (–) ve $\Rightarrow$ No solution.]

69. $\int_3^{x^3} e^{t^2}\,dt = \int_3^1 e^{t^2}\,dt + \int_1^{x^3} e^{t^2}\,dt = \int_1^{x^3} e^{t^2}\,dt - \int_1^3 e^{t^2}\,dt = g(x^3) - g(3)$

70. $I = \int_0^{\theta} \ln\left(1 + \tan\theta\tan(\theta - x)\right)dx$

$I = \int_0^{\theta} \ln\left(1 + \frac{\tan\theta(\tan\theta - \tan x)}{1 + \tan\theta\tan x}\right)dx = \int_0^{\theta} \ln\left(\frac{1 + \tan^2\theta}{1 + \tan\theta\tan x}\right)dx$

$I = \int_0^{\theta} \ln(1 + \tan^2\theta)\,dx - \int_0^{\theta} \ln\left(1 + \tan\theta\tan x\right)dx$

$I = 2\theta\, \ln \sec\theta - I$

$2I = 2\theta\, \ln \sec\theta \quad \Rightarrow \quad I = \theta\, \ln \sec\theta$

71. We have $r + s = -\frac{a_1}{a_2}$; $rs = \frac{a_0}{a_2}$

Simplifying the R.H.S. of the equation, we have

$$a_0\left(1-\frac{x}{r}\right)\left(1-\frac{x}{s}\right) = a_0\left(1-\frac{x}{r}-\frac{x}{s}+\frac{1}{rs}x^2\right) = a_0\left(1-\left(\frac{r+s}{rs}\right)x+\frac{1}{rs}x^2\right)$$

$$= = a_0 + a_1x + a_2x^2$$

Thus, the answer is (A) for all values of x, $a_0 \neq 0$]

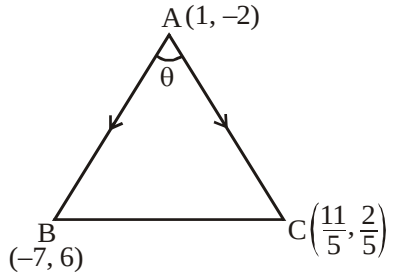
72. $\overrightarrow{AB} = \vec{V}_1 = -8\hat{i}+8\hat{j} = 8(-\hat{i}+\hat{j})$

$\overrightarrow{AC} = \vec{V}_2 = \left(\frac{11}{5}-1\right)\hat{i}+\left(\frac{2}{5}+2\right)\hat{j}$

$= \frac{6}{5}\hat{i}+\frac{12}{5}\hat{j} = \frac{6}{5}(\hat{i}+2\hat{j})$

$\cos\theta = \frac{(-\hat{i}+\hat{j})\cdot(\hat{i}+2\hat{j})}{(\sqrt{2})(\sqrt{5})} = \frac{-1+2}{\sqrt{10}} = \frac{1}{\sqrt{10}} = \tan\theta = 3$

$\Rightarrow$ (A)]



73. $y = Ax^2$; $y^2 + 3 = x^2 + 4y$; $A > 0$

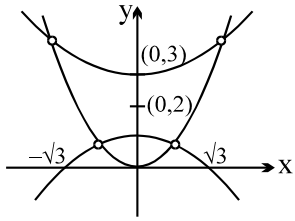
Now $y^2 - 4y = x^2 - 3$

$(y-2)^2 = x^2 + 1$

$(y-2)^2 - x^2 = 1$

if $x = 0$ $y - 2 = 1$ or -1 $\Rightarrow$ $y = 3$ or 1

Hence the two graphs of $y = Ax^2$ $(A > 0)$



and the hyperbola $(y-2)^2 - x = 1$ are as shown which can intersects in 4 points.]

74. $a^2 + b^2 + c^2 + ab + bc + ca \leq 0$

$(a+b)^2 + (b+c)^2 + (c+a)^2 \leq 0$

$\therefore$ $a + b = 0$

$b + c = 0$ $\quad\quad$ $a = b = c = 0$

$c + a = 0$

Now, $\begin{vmatrix} 4 & 0 & 1 \\ 1 & 4 & 0 \\ 0 & 1 & 4 \end{vmatrix} = 4(16-0) + 1(1-0) = 65$

75. using LMVT in [0, 1]

$f'(c) = \frac{f(1)-f(0)}{1-0}$ for some $c \in (0, 1)$

$e^{c^2} = \frac{f(1)-f(0)}{1}$; $f(1) - 10 =$ for some $c \in (0, 1)$

$1 < f(1) - 10 < e$; $\quad\quad$ $11 < f(1) < 10 + e$

$\therefore$ $A = 11$; $\quad\quad$ $B = 10 + e$ $\Rightarrow$ $A - B = 1 - e$

76. We have

$\frac{dV}{dt} = 2 \Rightarrow \frac{d}{dt}\left(\frac{1}{3}\pi r^3\right) = 2$ [Here r = h, as □ = 45°]

$\Rightarrow \pi r^2 \frac{dr}{dt} = 2 \Rightarrow \frac{dr}{dt} = \frac{2}{\pi r^2}$

Now, perimeter = $2\pi r$ = p(let)

$\Rightarrow \frac{d}{dt}(2\pi r) = 2\pi\left(\frac{2}{\pi r^2}\right) = \frac{4}{r^2}$ .... (1)

When h = 2 meters $\Rightarrow$ r = 2 meters

Hence $\frac{dp}{dt} = \frac{4}{4} = 1$ m/sec.]

77. Let $S = \sum_{r=1}^{\infty} \frac{r+2}{2^{r+1}\cdot r\cdot(r+1)} = \sum_{r=1}^{\infty} \frac{2(r+1)-r}{2^{r+1}\cdot r\cdot(r+1)} = \sum_{r=1}^{\infty} \frac{1}{2^{r+1}}\left(\frac{2}{r} - \frac{1}{r+1}\right)$

$= \sum_{r=1}^{\infty}\left(\frac{1}{2^r\cdot r} - \frac{1}{2^{r+1}(r+1)}\right)$

$= \frac{1-a\pi}{b\pi^2}$

$= \lim_{n\to\infty}\left[\left(\frac{1}{2^1\cdot 1} - \frac{1}{2^2\cdot 2}\right) + \left(\frac{1}{2^2\cdot 2} - \frac{1}{2^3\cdot 3}\right)\right] + \left(\frac{1}{2^3\cdot 3} - \frac{1}{2^4\cdot 4}\right) + ... + \left(\frac{1}{2^n\cdot n} - \frac{1}{2^{n+1}\cdot(n+1)}\right)$

$= \lim_{n\to\infty}\left(\frac{1}{2} - \frac{1}{2^{n+1}\cdot 9n+1}\right)$

$\therefore \quad S = \frac{1}{2}$

Hence $S^{-1} = 2$

78. Let the probability of the faces 1, 3, 5 or 6 be p for each face.
Hence probability of each of the faces 2 or 4 is 3p

$\therefore \quad 4p + 6p = 1 \quad \Rightarrow \quad p = \frac{1}{10}$

$\therefore \quad P(1) = P(3) = P(5) = P(6) = \frac{1}{10}$; $P(2) = P(4) = \frac{3}{10}$

P(total of 7 with a draw of dice) = P(16, 61, 25, 52, 43, 34)

$= 2\left(\frac{1}{10}\cdot\frac{1}{10}\right) + 2\left(\frac{3}{10}\cdot\frac{1}{10}\right) + 2\left(\frac{3}{10}\cdot\frac{1}{10}\right)$

$= \frac{2+6+6}{100} = \frac{14}{100} = \frac{7}{50}$

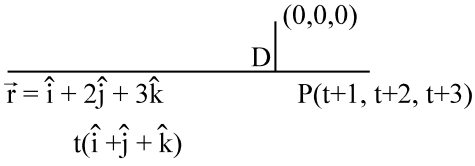
79. $D^2 = f(t) = (t+1)^2 + (t+2)^2 + (t+3)^2$

$f'(t) = 2[(1+t) + 2 + t + 3 + t]$

$f'(t) = 0 \quad \Rightarrow \quad t = -2$

hence $D^2(min) = 1 + 0 + 1 = 2$

$D_{min} = \sqrt{2}$



80. We have $|z - (1+i)|^2 = 2$

$\Rightarrow \quad (x-1)^2 + (y-1)^2 = 2$ (Put z = x + iy)

$\Rightarrow \quad x^2 + y^2 = 2(x+y)$ .....(1)

Let $\omega = h + ik = \frac{2}{z} = \frac{2}{x+iy} = \frac{2(x-iy)}{x^2+y^2}$, so

$$h = \frac{2x}{x^2+y^2}\ ,\ k = \frac{-2y}{x^2+y^2}$$

$\Rightarrow$ $h - k = \frac{2(x+y)}{x^2+y^2} = 1$ (from equation (1))

$\therefore$ Locus of the point $\omega(h, k)$ will be $x - y = 1$

81. If $\theta \in \left(\frac{3\pi}{2}, 2\pi\right)$, then $\tan\theta < 0$

Given $a^{\tan\theta} > 1$ $\Rightarrow$ $0 < a < 1$

and $b^{\tan\theta} > 1$ $\Rightarrow$ $0 < b < 1$

Also $a^{\tan\theta} > b^{\tan\theta}$ (given) $\Rightarrow$ $b > a$

Hence $a < b < 1$

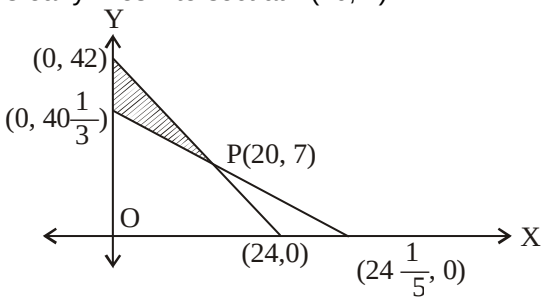
82. We have $(x^2 + a|x| + a + 1)(x^2 + (a+1)|x| + a) = 0$

Clearly for $a > 0$,

$x^2 + a|x| + a + 1 > 0 \quad \forall x \in R$ and $x^2 + (a+1)|x| + a > 0 \ \forall x \in R$

Hence the given equation has no real root if $a \in (0, \infty)$

83. Clearly lines intersect at P(20, 7)



$\therefore$ Area of shaded region $= \frac{1}{2}\left(42 - 40\frac{1}{3}\right)20 = \frac{1}{2}\left(\frac{5}{3}\right)20 = \frac{50}{3}$ (square units)

84. Given that $p_1, q_1, p_2 q_2$ are in A.P.

$\therefore$ $(p_2 - p_1)^2 = (q_2 - q_1)^2$

$\Rightarrow$ $(p_2 + p_1)^2 - 4p_1p_2 = (q_2 + q_1)^2 - 4q_1q_2$

$\Rightarrow$ $\left(\frac{-b}{a}\right)^2 - 4\left(\frac{c}{a}\right) = \left(\frac{-b}{c}\right)^2 - 4\left(\frac{a}{c}\right)$

$\Rightarrow$ $\frac{b^2 - 4ac}{a^2} = \frac{b^2 - 4ac}{c^2}$

Since $b^2 - 4ac$ is the discriminant of both the equations and roots are different

$\therefore$ $b^2 - 4ac > 0$

$\therefore$ $a^2 = c^2$

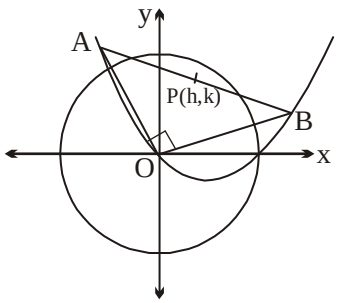
$\Rightarrow$ $a = c$ (Not possible because two quadratic equations become identical)

or $a = -c \Rightarrow a + c - 1 = -1$

85. Equation of chord of the circle with mid point (h, k) is

$hx + ky = h^2 + k^2$

On homogenising the parabola, we get



$$x^2 - \frac{(x+y)(hx+ky)}{h^2+k^2} = 0$$

$\Rightarrow$ $(h^2 + k^2)x^2 - (x + y)(hx + ky) = 0$

Now coefficient of x^2 + coefficient of $y^2 = 0$

$\therefore$ $h^2 + k^2 - h - k = 0$

$\Rightarrow$ $x^2 + y^2 - x - y = 0$ $\Rightarrow$ circle with centre $\left(\frac{1}{2},\frac{1}{2}\right)$ and radius = $\sqrt{\frac{1}{4}+\frac{1}{4}} = \frac{1}{\sqrt{2}}$

86. Since root(s) are real and distinct, so

$36 - 4l > 0 \Rightarrow 9 > l$

If x_1, x_2 be roots then $x_1 + x_2 = -6$ and $x_1x_2 = l$

Also $|x_1 - x_2| \le 4$ i.e. $(x_1 - x_2)^2 \le 16$

or $(x_1 + x_2)^2 - 4x_1x_2 \le 16$

$\Rightarrow$ $36 - 4l \le 16$ or $5 \le l$

$\therefore$ $5 \le l < 9$ i.e. $l \in [5, 9)$

87. If number of sides is n, then

total number of diagonals of a convex polygon = ${}^nC_2 - n = 44$ (given)

$\Rightarrow$ $n(n - 1) - 2n = 88$ $\Rightarrow$ $n^2 - 3n - 88 = 0$ $\Rightarrow$ $(n - 11)(n + 8) = 0$

$\Rightarrow$ $n = 11$ $\Rightarrow$ undecagon

88. (i) T S S T S S T S S

(ii) S T S S T S S T S

(iii) S S T S S T S S T

Hence $3 \cdot (3!)6! = (18)6!$

$k = 18$

89. Required number of ways = (number of ways in which 16 players can be divided in 8 couples) – (number of ways when S_1 and S_2 are in the same group)

$$= \frac{16!}{2^8\cdot 8!} - \frac{(14)!}{2^7\cdot 7!} = \frac{16\cdot 15\cdot 14!}{2\cdot 2^7\cdot 8\cdot 7!} - \frac{(14)!}{2^7\cdot 7!}$$

$$= \frac{(14)!}{2^7\cdot 7!}\left[\frac{16\cdot 15}{16} - 1\right] = \frac{(14)\cdot(14)!}{14\cdot(6!)\cdot 2^6} = \frac{(14)!}{2^6\cdot 6!}$$ **Ans.**]

90. Let $z = x + iy$

$$E = z\bar{z} + (z-3)(\bar{z}-3) + (z-6i)(\bar{z}+6i)$$

$$= 3z\bar{z} - 3(z+\bar{z}) + 9 + 6(z-\bar{z})i + 36 \qquad (z+\bar{z} = 2x;\ z-\bar{z} = 2iy)$$

$= 3(x^2 + y^2) - 6x - 12y + 45$

$= 3[x^2 + y^2 - 2x - 4y + 15]$

$E = 3[(x - 1)^2 + (y - 2)^2 + 10]$

Hence $E_{min} = 30$ when $x = 1$ and $y = 2$ i.e. $z = 1 + 2i$]